

INSTITUTIONAL DEFI INFRASTRUCTURE

VeritasVault

A dual-brand institutional DeFi platform — market data, risk analytics, treasury and governance — engineered so that every action is provable, auditable and survivable under regulatory and adversarial scrutiny.

DOCUMENT

Investor Overview

STAGE

Pre-seed · Pre-revenue

PREPARED

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An institutional data and risk backbone for on-chain finance

The thesis

Institutional capital cannot enter DeFi on retail-grade infrastructure. What is missing is not another exchange or yield venue — it is the provable layer underneath: versioned market data, enforceable risk policy, and an audit trail a regulator will accept.

The product

A dual-brand platform — an institutional tier and a private-client tier — designed around a Black-Litterman allocation engine with ML-enhanced covariance estimation, over a seven-domain architecture spanning market data, risk, settlement, governance and AI/ML.

Where it stands

Deployed and publicly reachable at veritasvault.net — a Next.js 15 application with 36 dashboard views and 44 API routes, over a .NET 9 market-data core, across eleven repositories. Pre-revenue, and currently between development pushes.

1,816

TYPESCRIPT / REACT FILES

Across the web platform, docs portal and game suite

36

DASHBOARD VIEWS

Portfolio, risk, treasury, voting, strategies, liquidity

383

SPECIFICATION DOCUMENTS

Domain, contract and security specs under version control

11

REPOSITORIES

Web, backend, chain services, contracts, laC, docs

WHAT THIS DOCUMENT DELIBERATELY OMITS

Market sizing, financial projections and the terms of the raise are not in this document. Every figure here is measured from the codebase; anything that would have to be estimated is discussed directly rather than asserted on a slide.

DeFi was not built to be audited

Every structural property that made decentralised finance fast to build is the same property that makes it unusable for a regulated allocator.

I

Data without provenance

Prices, curves and surfaces arrive unversioned and unattributed. When a valuation is challenged months later, there is no defensible record of what the system knew, when it knew it, or which revision drove the trade.

II

Risk policy that is advisory

Limits, KYC/AML checks and exposure rules live in dashboards and off-chain processes rather than being enforced at the point of execution. Controls that can be bypassed are not controls.

III

Compliance retrofitted late

Protocols are built first and made compliant afterwards. The result is expensive remediation, jurisdictional lock-out, and audit trails assembled after the fact from logs that were never designed as evidence.

THE CONSEQUENCE

Institutions are structurally excluded. Not because the yield is unattractive, but because no custodian, auditor or risk committee can sign off on infrastructure that cannot prove its own history. The blocker is evidentiary, not economic.

Who feels this

- **Asset managers** needing defensible marks and position limits
- **Custodians** requiring multi-signature and time-locked control paths
- **Compliance teams** obliged to produce jurisdiction-specific reporting
- **Auditors** who must verify state independently, not take it on trust

Provable by construction, not by policy

VeritasVault inverts the usual order: the evidence layer is designed first, and every feature above it inherits auditability rather than bolting it on.

Immutability & audit

Every entity is versioned and every state transition emits a signed, append-only record. Nothing is silently overwritten.

Regulatory-first

KYC/AML, position limits and reporting are domain contracts enforced at execution — not dashboards observed after it.

Zero trust

Every interface is authenticated, authorised and monitored by default, including internal domain boundaries.

Defense in depth

Layered controls, circuit breakers and multi-signature paths so that a single compromise is contained, not fatal.

ARCHITECTURE THAT MAKES THIS ENFORCEABLE

- **Clean / hexagonal layering** — the domain sits at the centre and depends on nothing; infrastructure and API are adapters pointing inward.
- **Explicit contracts per domain** — each of the seven domains is independently auditable, upgradeable and replaceable without cascading change.
- **CQRS with mediated handlers** — read and write paths separate, so audit and query load never contend with settlement correctness.
- **Versioned schemas and events** — schema evolution is a first-class concern, allowing historical records to remain readable and provable.

If you can't prove it, you can't do it.
*If you can't audit it, **it doesn't exist.***

— VeritasVault domain design principle

Two front doors, one platform

A single codebase serves an institutional tier and a retail tier, with separate authentication, navigation, theming and feature surfaces.

Corporate tier

INSTITUTIONAL · 19 DASHBOARD VIEWS

Treasury, portfolio and model-portfolio management, on-chain and off-chain analytics, governance voting, risk assessment, flash loans, Tezos liquidity, watchlists, event tracking and an admin console — plus a full marketing, pricing, compliance and security-whitepaper site.

Standard tier

RETAIL · 17 DASHBOARD VIEWS

Liquidity pool discovery and detail, strategy creation and management, risk assessment, analytics with custom dimensions and funnel reporting, profile and settings — with a guided demo mode for prospects who have not yet connected real capital.

REPRESENTATIVE DASHBOARD SURFACE

Portfolio Holdings & allocation	Treasury Reserves & flows	Risk assessment Exposure scoring	Strategies Create, compare, deploy	Liquidity pools Discovery & detail
Voting Governance proposals	On / off-chain Split analytics	Performance Attribution	AI analytics Model insights	Flash loans Execution monitor

ALSO IMPLEMENTED

Five AI endpoints — pool analysis, strategy generation, risk assessment and visualisation generation — alongside authentication with social login, a fourteen-migration Postgres schema, and live integrations to DefiLlama, Goldsky and CoinGecko.

Black-Litterman, enhanced where it is weakest

The allocation framework institutional asset management already trusts — ported to digital assets, and improved at its most fragile input.

Why this is the wedge

Black-Litterman is the language institutional allocators already speak. Arriving with it removes the hardest part of an enterprise sale — arguing that the method deserves trust — and reframes the conversation as implementation rather than principle.

The technical contribution

The model's most fragile input is the covariance matrix. The specification applies ML shrinkage estimation — evaluated against Frobenius, KL-divergence and spectral loss, with ensemble architecture and uncertainty quantification — reducing noise without disturbing the Bayesian framework.

Auditability inside the model

Each view carries an author, rationale, evidence, confidence and expiry; each result reports per-view impact, weight change and tilt from equilibrium. Allocation is attributable by construction — the platform's thesis applied to the portfolio itself.

45

SPECIFICATION DOCUMENTS

Model, views, validation, ML covariance

3

LOSS FUNCTIONS EVALUATED

Frobenius, KL-divergence, spectral

Status — stated precisely

Specification	COMPLETE	45 documents
Data contracts	COMPLETE	full input, view and result DTOs
Presentation layer	SHELL	renders sample output
Solver & ML estimator	NOT BUILT	the core Phase 1 deliverable

WHAT THIS IS, AND IS NOT

The design work is done and the contracts are settled; the computation is not written. That is a well-scoped engineering task against a finished specification — which is a very different risk from an unsolved research problem, and should be diligenced as such.

Seven domains, one dependency rule

Each domain owns its models, contracts and events. Dependencies point inward toward pure domain logic — no domain reaches around another.

1 · Core Infrastructure

Consensus and finality, chain indexing, gas policy, rate limiting, fork and upgrade management.

ConsensusManager · ChainIndexer · ForkManager

2 · Risk, Compliance & Audit

Real-time and historical risk models, KYC/AML enforcement, tamper-proof audit logging.

RiskModel · ComplianceManager · AuditLogger

3 · Asset, Trading & Settlement

Portfolios, asset lifecycle, AMM liquidity, deterministic order matching and atomic settlement.

Portfolio · OrderBook · SettlementController

4 · Integration & Analytics

Cross-chain bridging, consensus price oracles, API perimeter, analytics pipeline and data lake.

Bridge · PriceOracle · AnalyticsEngine

5 · Governance, Ops & Custody

On-chain proposals and voting, versioned parameter store, treasury, insurance fund, time-locked upgrades, escrow and dispute resolution.

GovernanceController · ParameterStore · EscrowController

6 · AI / ML Governance

Model registry and versioning, shadow and canary deployment, continuous fairness and drift monitoring, operator staking and slashing.

GlobalModelRegistry · ContinuousFairnessController

7 · Integration Gateway

API key lifecycle and rate limiting, protocol adapter and bot management, integration orchestration, event and notification delivery.

APIGateway · AdapterManager · MessageBus

WHY THIS MATTERS COMMERCIALLY

Modularity is the go-to-market. A counterparty can adopt the market-data and audit layer without taking on trading or governance — and each domain can be security-reviewed and certified on its own timeline rather than as one monolithic approval.

Eleven repositories, measured

File counts taken directly from each repository's git index — not from planning documents.

REPOSITORY	ROLE	SCALE	LAST PUSH
vv-landing	Dual-brand web platform — marketing, auth, dashboards, analytics, AI endpoints	933 TS/TSX · ~87k LOC · 93 pages · 44 API routes · 455 components	Jan 2026
vv	.NET 9 market-data core — domain, CQRS, Cosmos persistence, event publication	157 C# files · 12.1k LOC · 383 specification documents	May 2026
vv-game-suite	WebGL / Phaser framework — risk-liquidity puzzles and governance simulations	483 TS/TSX · 45 pages and routes	Jul 2025
vv-docs	Docusaurus documentation portal — API reference, tutorials, design guidelines	231 TS/TSX · 122 MD/MDX documents ARCHIVED	Feb 2026
vv-chain-services	Blockchain microservices — Ethereum, Tezos, Postgres, Fastify	90 files · 12 Python · 5 C# · 4 test files	Jun 2025
vv-iac	Azure infrastructure as code with what-if validation and gated promotion	42 Bicep templates · 64 files	May 2025
vv-chain	Cross-chain risk intelligence — on-chain contracts, TVL and protocol health	10 Solidity contracts	May 2025
vv-auth · vv-auth-frontend-demo	Authentication service and reference front end	109 TS/TSX across both	Jul 2025
cognitive-mesh-nexus	AI mesh integration, plus shared developer tooling in vv-dev-tools	60 TS/TSX · 32k of PowerShell tooling	Jul 2025

Four operating systems VeritasVault does not have to build

Infrastructure already running across our wider portfolio and reusable directly by VeritasVault — and, candidly, where most recent engineering activity has sat.

Sluice

AI GATEWAY · PYTHON · LIVE

OpenAI-compatible gateway on Azure Container Apps with LiteLLM routing, model abstraction and cost control. VeritasVault's five AI endpoints route through a governed layer instead of binding to one vendor.

Docket

AI COST OPS · LIVE

LLM spend tracking, usage analytics and cost optimisation. Makes the unit economics of AI-assisted allocation measurable rather than assumed — a question every investor asks of an AI-heavy product.

Cognitive Mesh

AGENT PLATFORM · .NET · LIVE

Enterprise agent and LLM platform with RBAC, audit logging, Azure OpenAI and RAG. The governance scaffolding VeritasVault's AI/ML domain specifies, already implemented in an adjacent product.

Baton

TASK GRAPH · TYPESCRIPT · LIVE

Shared human-and-agent task graph with a Kanban interface, MCP server and bidirectional YAML sync. The coordination layer that makes a small team running many agents tractable.

WHY THIS MATTERS TO AN INVESTOR

An AI-heavy financial product normally has to build model routing, cost control, agent governance and audit from scratch — expensive, and squarely off the critical path. Here they already exist, are running, and carry no additional build cost.

THE OTHER READING — STATED PLAINLY

All four were pushed within the last week; VeritasVault's last substantial push was January. That is the honest evidence on the focus question, and it is the right thing to interrogate. The case for the raise is precisely that it buys concentrated attention on VeritasVault.

What is built, what is specified, what is next

An honest delineation across the whole estate.

CAPABILITY	STATUS	EVIDENCE
Production deployment	LIVE	Serving on Vercel at veritasvault.net — marketing, pricing, solutions, whitepaper and demo routes publicly reachable; dashboards correctly behind authentication.
Dual-brand web application	BUILT	93 pages across corporate and standard tiers, 455 components, unified header/footer, theming, version selection and guided demo mode.
Dashboards & analytics	BUILT	36 dashboard views — portfolio, treasury, risk, voting, strategies, pools, performance, on/off-chain — with Recharts visualisation and funnel reporting.
Auth, data & API layer	BUILT	44 API routes, NextAuth with GitHub and social login, Supabase Postgres with 14 sequential migrations plus seed data.
Market data core (.NET)	BUILT	Six versioned entity types over BaseMarketData ; Cosmos DB repository partitioned on <code>/assetId</code> with optimistic concurrency; MediatR CQRS; Event Grid and MassTransit publication.
External data integrations	BUILT	DefiLlama TVL, Goldsky protocol indexing and CoinGecko market data, each with retry, exponential backoff and explicit fallback handling.
Black-Litterman engine	SPECIFIED	45 documents and complete data contracts; the UI renders sample output. The solver and ML covariance estimator are not implemented — this is the headline Phase 1 build.
AI features	PARTIAL	Five endpoints live — pool analysis, strategy generation, risk assessment, visualisation. Model governance (registry, fairness, drift) remains specified only.
Automated testing	WEAK	Backend at 11.5% line coverage across 15 test files; the web platform has no test files at all . The single largest engineering gap.
Risk engine, governance, bridging	SPECIFIED	Contract, event and security specification across 383 documents; UI surfaces exist, but the enforcing backend domains are not yet implemented.

A conservative stack for a conservative buyer

Institutional buyers reward boring, supportable technology choices. The stack is deliberately mainstream, managed and auditable.

WEB PLATFORM

Next.js 15.3 React 19 TypeScript 5.8 Tailwind Radix UI Recharts

IDENTITY & DATA

NextAuth Supabase PostgreSQL Zod

BACKEND SERVICES

.NET 9 ASP.NET Core MediatR / CQRS Azure Functions Fastify

INFRASTRUCTURE & CHAIN

Cosmos DB Event Grid MassTransit Bicep Solidity Tezos Data Lake

Solid chips are implemented today; outlined chips are specified and planned.

Deliberately mainstream

No bespoke consensus client, no exotic runtime, no dependency on a single chain's tooling. This materially reduces both audit surface and the difficulty of hiring engineers who can maintain it.

Data design

Composite document identity encodes data type, asset class, asset, region, date, document type, schema version and entity version. A record is self-describing and independently verifiable without reference to external state.

Engineering discipline

Specification-first development with contract documents preceding implementation; conventional commits, pre-commit hooks, CI workflows, duplication detection and coverage reporting wired into the repositories.

Where the defensibility actually sits

1 · The allocation engine

Black-Litterman with ML-enhanced covariance estimation and view-level attribution — the language institutional allocators already speak, specified across 45 documents with contracts settled. This is the wedge, not the architecture.

2 · Compliance as architecture

Most entrants treat regulation as a feature request. Here, policy enforcement points, audit emission and jurisdictional pluggability are structural — which is precisely what shortens an institutional procurement cycle.

3 · Governed AI/ML, not bolted-on AI

AI endpoints are live, and the governance layer around them — registry, shadow and canary deployment, fairness and drift monitoring, operator staking — is fully specified. As model-governance regulation tightens, this becomes an entry requirement.

Adjacent approaches and their limits

APPROACH	STRUCTURAL LIMITATION
Retail DeFi protocols	No audit trail designed as evidence; compliance bolted on post-hoc.
Traditional data vendors	Strong provenance, but no on-chain settlement or programmable policy enforcement.
Custody-only providers	Solve asset safety, not valuation defensibility or risk-policy enforcement.
In-house institutional builds	Expensive, slow, and rebuilt independently at every institution.

COMPETITIVE LANDSCAPE

Named competitor benchmarking and pricing comparison are provided on request as part of diligence. With no customers yet, we have no win/loss evidence — and would rather say so than present the category analysis above as though it were proven in the field.

Phased delivery, gated on evidence

Each phase ends at a demonstrable gate rather than a date — security review, audit logging and integration proof are release criteria, not follow-up work.

Phase 1 IN PROGRESS	Harden what exists Test coverage across the web platform and the .NET core, wiring the dashboards to the market-data backend rather than fallback data, and hardening the existing 44 API routes. This converts a demonstrable product into a sellable one.
Phase 2 NEXT	Live data, risk enforcement and gateway Low-latency live-value container, price oracle consensus and anomaly detection, baseline risk model and compliance enforcement behind the existing risk-assessment UI, and the API gateway with key lifecycle and rate limiting.
Phase 3 PLANNED	Archival, analytics and asset lifecycle Data Lake cold-storage tiering with backfill, analytics for operational and compliance reporting, and auditable trade handling behind the portfolio surfaces.
Phase 4 SPECIFIED	Governance, custody and AI/ML governance On-chain voting behind the existing UI, versioned parameter store, treasury and insurance fund, time-locked upgrades, and the governed model registry.

GO-TO-MARKET SEQUENCING · AND SCHEDULE CANDOUR

Private investors first, institutions second. A small private-client cohort validates the allocation engine, generates reference customers and produces revenue against a far shorter sales cycle — while the compliance surfaces an institutional buyer requires are completed. The corporate tier is the larger prize, not the first move.

The original Phase 1 plan targeted an MVP gate in mid-2025; that date was missed, and the last substantial web push was January 2026. Dates are omitted until resourcing is settled — delivery velocity, not architecture quality, is the primary diligence question.

Where the platform can expand

Eight expansion vectors identified in the platform's strategic assessment, ordered by proximity to what is already built.

Institutional onboarding

Enterprise custody, multi-signature governance and compliance reporting open access to institutional capital and liquidity.

Regulatory adoption

Compliance-by-design becomes a competitive advantage precisely as regulated markets formalise digital-asset rules.

Audit & proofs as a service

Exportable, cryptographically verifiable audit data is a standalone revenue line independent of trading volume.

AI/ML and DeFi fusion

Governed models applied to portfolio optimisation, anomaly detection and bridge security — auditable by construction.

Cross-chain growth

Chain adapters and bridging extend reach beyond a single ecosystem and reduce dependence on any one chain's economics.

Composable ecosystem

Standardised interfaces allow third-party strategy plugins, protocol adapters and model components — network effects without core headcount.

Pluggable policy modules

Jurisdiction-specific compliance and risk models let one codebase serve materially different regulatory regimes.

Automated incident response

Coordinated circuit breakers and self-healing reduce operational risk — a direct procurement argument for regulated buyers.

ON SIZING

This slide describes the direction of expansion, not a sized market. Addressable market, segment prioritisation and revenue modelling are worked through directly rather than presented as a headline number we would then have to defend.

Known risks and how they are addressed

Drawn from the platform's own strategic assessment and from direct inspection of the codebase.

RISK	EXPOSURE	MITIGATION
Test coverage	The web platform has no automated tests; the backend sits at 11.5% line coverage.	Coverage is the Phase 1 exit gate, starting with API routes and the risk surfaces.
Focus & continuity	Last substantial VeritasVault push was January 2026, while four sister projects were updated this week.	The raise buys concentrated attention; sister infrastructure is reused, not rebuilt.
Front-end / back-end integration	Dashboards partly run on fallback and demo data rather than the .NET market-data core.	Wire the existing UI to live services as the first Phase 1 deliverable.
Estate fragmentation	Eleven repositories with divergent activity raise coordination and drift cost.	Consolidate or archive dormant repos; one release process across the active core.
Execution & scope	Seven-domain ambition against a small team; Phase 1 slipped against plan.	Strict phase gating; treat scope reduction as the default lever.
Regulatory uncertainty	Rules across jurisdictions are unsettled and may force compliance redesign.	Pluggable jurisdiction-specific policy modules; standing regulatory monitoring.
Bridge & adapter exploits	Cross-chain components are the sector's highest-value attack surface.	Additional review, dedicated monitoring, circuit breakers, staged rollout.
Key-person concentration	Engineering depth currently rests with a single technical founder.	Specification-first practice keeps the system legible to incoming engineers; first hires are the priority on close.

What funding would buy

The estate is unusually complete for its stage. Capital converts it from demonstrable to sellable.

Harden and connect

Bring the web platform under test, wire the 36 dashboard views to the .NET market-data core instead of fallback data, and harden the existing API surface. Highest-value use of the first tranche by a wide margin.

Security control harmonisation

Audit and consolidate overlapping controls across domains into a unified policy framework with clear boundaries and ownership.

Regulatory readiness programme

Jurisdiction-specific compliance packages, regulatory monitoring capability, and early relationships with regulators in target markets.

Institutional onboarding

Enterprise documentation, reporting tooling and a streamlined onboarding path — the work that converts architectural credibility into signed counterparties.

Also funded

- **Cross-domain monitoring** — unified health checks, watchdogs and incident procedures spanning domain boundaries
- **Operational excellence** — attack simulations, automated cross-domain testing, runbooks for every critical process
- **Ecosystem development** — SDKs, developer documentation and a security review process for third-party contributions
- **Cross-chain security** — hardened treatment of adapters, oracles and bridges as the highest-risk surfaces
- **Estate consolidation** — archive dormant repositories, unify release and CI across the active core

TERMS

Amount · Instrument · Valuation · Runway · Milestone and hiring plan

Discussed directly rather than published in a document that will be forwarded. The priorities above are the substance of the request; the structure around them is a conversation.

The hard part of institutional DeFi is the evidence layer.

VeritasVault has designed it — and built on it.

Built

A dual-brand Next.js platform with 36 dashboard views and 44 API routes, on a .NET 9 market-data core with versioned entities and event publication.

Designed

383 specification documents across seven domains — contracts, events, security threat models and implementation guidance.

Needed

Capital and engineering capacity to test, connect and harden what already exists — then to open the specified domains in sequence.

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PLATFORM

veritasvault.net

DOCUMENT

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